DILEEPA PITAWELA

DATA ANALYST / DATA ENGINEER / DATA SCIENTIST
AI/ML ENGINEER / APPLIED SCIENTIST
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Adelaide, SA, Australia | Open to relocation | Available from April 2026

ABOUT ME

I'm currently completing my Ph.D. in Computer Science at the University of Adelaide and hold a Master's degree in Data Science. I have 3+ years of data/software engineering experience and over 3.5 years of research experience. My expertise includes computer vision, deep learning, data wrangling, experimental design, and solution engineering. I am seeking a Data or AI/MLE role to develop machine learning systems for real-world applications. As I transition from PhD to industry, I am also open to early-career or graduate opportunities.

TECHNICAL SKILLS

AI & ML: Computer Vision, Deep Learning, Generative AI, Human-AI Collaboration, NLP
Frameworks: PyTorch, TensorFlow, LangChain, Keras, RAG, Hadoop
Programming: Python, C#(.net), Java, Angular, Laravel, JavaScript, TypeScript
Cloud & CI/CD: AWS (S3, Lambda, CloudFront, Bedrock, IAM), Azure (DevOps, Storage), Git
Analytics & Visualisation: pandas, scikit-learn, matplotlib, seaborn, PowerBI, Excel
Databases: MSSQL, MySQL, MongoDB, Ontology, Vector DB, Redis caching

PROFESSIONAL EXPERIENCE

Doctoral Researcher at The University of Adelaide **Oct 2022 - Present**

- Working on latent feature disentanglement in computer vision to enable semantically meaningful data generation with generative models.
- Exploring interpretable and controllable latent representations for vision tasks.
- Studying human-AI collaboration for AI-assisted complex decision-making.

Data Engineer at The University of Adelaide **Aug 2022 - Aug 2023**

- Developed a web simulator and a novel dataset for Web Vision Language Navigation (VLN).
- Contributed to multimodal web navigation models integrating vision and text.

Data/ML Engineer Intern at The Centre for Research on Engineering Software Technologies **Nov 2021 - Feb 2022**

- Applied ML for causal analysis of software vulnerabilities before and after deployment.

Research Engineer Intern at The Centre for Research on Engineering Software Technologies **Nov 2020 - Feb 2021**

- Simulated and analysed software security threats in multi-agent systems using Java JADE.

Software Engineer at 99X **Jul 2018 - Jul 2020**

- Developed core modules of the Task Manager product within the Adra Suite, used by 18,000+ clients across 20 countries.
- Collaborated with global teams (Norway and the US) on feature development and led client discussions to deliver features and shape product roadmaps.

Software Engineer Intern at Virtusa **Dec 2016 - Jun 2017**

- Developed core modules of the in-house process automation platform "Tempo", aimed at streamlining routine workflows using Java Spring Boot, JavaFX, and MSSQL.

EDUCATION

PhD in Computer Science at University of Adelaide, Australia **Oct 2022 - Present**
MSc in Data Science at University of Adelaide, Australia **Jul 2020 - Jul 2022**
MSc in Project Management at University of Bedfordshire, UK **Sep 2022 - Sep 2023**
BSc (Hons) in Information Technology at University of Moratuwa, Sri Lanka **Apr 2014 - Jul 2018**

NOTABLE PUBLICATIONS

- CVPR 2025 - Contrastive Learning for Ordinal Classification
- AAAI 2024 - Vision-and-Language Navigation on Websites
- IJCNN 2022 - Document-aware Positional Encoding for Multi-Doc Summarization

Please see more on Google Scholar

NOTABLE PROJECTS

Order Preserving Data Augmentation **Oct 2024 - Present**
During PhD at The University of Adelaide

- Developed a latent image augmentation method using feature disentanglement and generative models to control the composition and proportions of features during image synthesis, enabling targeted generation of desired attributes.
- Implemented using VQ-VAE, Diffusion Models, PyTorch, Python

Ordinal Classification **Nov 2023 - Nov 2024**
During PhD at The University of Adelaide

- Proposed a novel representation learning algorithm that preserves ordinal relationships of data using learnable multiple margins, especially for ordinal datasets such as medical images or anomaly progression in products.
- Worked on developing deep learning architectures with contrastive learning.

AI in Healthcare During PhD at The University of Adelaide **Oct 2022 - Present**

- Developed multimodal models for dysplastic barrett's oesophagus detection on white light (WLI) and narrow band (NBI) imaging modalities.
- Worked on automating prompts for foundation video segmentation models.
- Contributed to optimizing human effort in interactive prompting workflows.

Human-AI Teaming During PhD at The University of Adelaide **Oct 2022 - Present**

- Proposed a framework that models human biases to enable AI-assisted decision-making at optimal intervention points.
- Designed algorithms for decision optimisation.

Web-Vision Language Navigation **Aug 2022 - Aug 2023**
During Data Engineer Tenure at The University of Adelaide

- Developed a web simulator to record user interactions and curate data for training Web-VLN models, using Python, Angular, Selenium and AWS.
- Integrated the simulator with the model to enable real-time web actions.

Task Manager During Software Engineer Tenure at 99X **Jul 2018 - Jul 2020**

- Delivered key product features with full ownership of end-to-end requirements.
- Led client meetings to gather requirements and demonstrate feature deliverables.
- Resolved technical debt to improve design and code quality to align with industry standards.
- Practiced Agile and used C#, .NET, MSSQL, Azure, JavaScript, and Git.

Please see more on LinkedIn

AWARDS & RECOGNITION

- **Global Citizens Scholarship**, The University of Adelaide, 2020
- **Golden Key International Honour Society Invitation**, 2021; Awarded to the top 15% of students at the university based on academic merit.
- **Adelaide Graduate Award**, University of Adelaide, 2022; In recognition for engagement in career development and volunteering beyond academics.
- **Higher Degree by Research (HDR) Scholarship**, The University of Adelaide, 2022–Present
- **Travel Grant**, The University of Adelaide, 2025

TRANSFERABLE SKILLS

- Problem Solving
- Adaptability
- Resilience
- Team Player
- Communication
- Engineering Thinking
- Continuous Learning
- Teaching & Mentoring
- Solution Designing
- Time & Resource Management

VOLUNTEERING

International Peer Mentor at the University of Adelaide **Jul 2021 - Dec 2023**

- Led mentorship sessions to share knowledge with commencing students.
- Hosted student engagement and cultural engagement events.

Webmaster at Golden Key International Honour Society **Jun 2021 - Oct 2022**

- Managed digital outreach and social media presence.

Organiser/Event Coordinator at University of Moratuwa **Jan 2018 - Jul 2018**

- Organised career and industry-bridging events.

REFERENCES

I would be pleased to provide references upon request.